



Spring 2026

EMCH792 – Optimal State Estimation

Homework 5

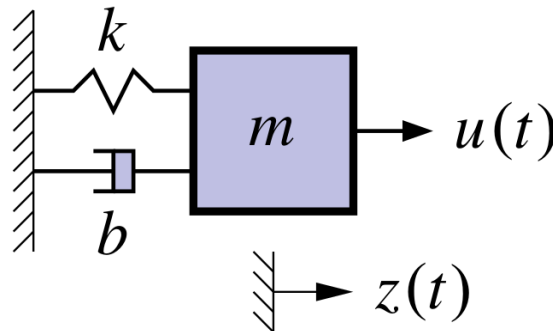
Submission deadline: Wednesday, April 1, 11:59pm EDT

- Submission only through Blackboard, email submissions will not be accepted.
- Make sure you properly upload the file(s) and that you can successfully view your submission on BB. Keep a copy of the submission receipt you get from BB.
- Late submissions will be graded with the policy described in the syllabus.
- For questions email tsucin@email.sc.edu and copy vitzilaos@sc.edu

Homework Instructions

In this homework you will learn how to implement a few generalizations of a Linear Kalman Filter in **MATLAB**. All the files that you will need for this homework are in the **ws** folder. You are expected to work inside this folder only. Open **MATLAB** and set the current folder to **ws**, and do the following:

- Start by opening the live script file “HW_5_manual.mlx” using **MATLAB**. This will guide you through the basic implementation of a Square-Root Kalman filter in **MATLAB**.
- Open the file “HW_5.m” and follow the instructions in the file to solve the following problem:



Consider the mass-spring-damper system shown in the picture above. The state of the system is the position and velocity of the mass. The process noise covariance is $Q = [0.005, 0; 0, 0.005]$. You have two sensors that can measure the position of the mass. The first one with an error with zero mean and variance of 0.025 and the second one with an error with zero mean and variance of 0.001. The second sensor, while more accurate, is prone to faulty measurements. All the data needed for this homework is in the “HW5_data.mat” file and is loaded at the beginning of the HW_5.m file. The true position and velocity of the mass are pos_true and vel_true respectively, the input to the system is u , $y1$ is the measurements of the first sensor and $y2$ the measurements of the second sensor.

1. For $m=1\text{kg}$, $b=2.5\text{Nsec/m}$ and $k=5.0\text{N/m}$ derive the discrete time system dynamics for $T_s = 0.1\text{sec}$.
2. Set the initial estimated state to $[0, 0]$, and the initial error covariance to $[1, 0; 0, 2]$. Design a Square-Root Kalman filter and simulate the system for all timesteps provided while collecting the estimated state.
3. For the Square-Root Filter, plot the true, estimated, and measured position. Plot the true and estimated velocity.
4. Set the initial estimated state to $[0, 0]$, and the initial error covariance to $[1, 0; 0, 2]$. Design a Sequential Kalman filter with measurement gating and simulate the system for all timesteps provided while collecting the estimated state. Use a chi critical for upper tail area of 0.005.
5. For the Sequential Filter, plot the true, estimated, and measured position. Plot the true and estimated velocity.

Submission Instructions

Your submission must include the following file:

- “HW_5.m” -> The edited MATLAB script with your solutions

Upload the file on Blackboard in a single submission.

Make sure that the scripts you submit can run. Do not copy the results that appear in the command window in your scripts and make sure that if you add any comments in your scripts that they follow the proper **MATLAB** syntax. **Scripts that do not run will not be graded.**

Grading Rubric

Problem	Points
1	5
2	35
3	10
4	40
5	10